Tutorial XV

21.4.2020

Q1. X_1, X_2 and X_3 is a random sample of size 3 from a population with mean μ and variance σ^2. $\Theta_1, \Theta_2, \Theta_3$ are the estimators used to estimate mean value μ, where

(i) $\Theta_1 = X_1 + X_2 - X_3$ (ii) $\Theta_2 = 2X_1 + 3X_2 - 4X_2$

(iii) $\Theta_3 = \frac{1}{3}(\lambda X_1 + X_2 + X_3)$

(a) Are Θ_1 and Θ_2 ube?

(b) Find λ such that Θ_3 is ube for μ.

(c) For this value of λ, is Θ_3 a consistent estimator?

(d) Which is the best estimator

SOLUTION
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Q1.

(a) $E[\theta_1] = E[X_1+X_2-X_3] = \mu$

$E[\theta_2] = E[2X_1+3X_2-4X_2] = \mu$ } Both are ube for μ (1)

(b) $E[\theta_3] = E[(\lambda X_1 + X_2 + X_3)/3] = \mu$ (2)

or $\lambda = 1$

(c) For $\lambda = 1$, $E[\theta_3] = E\left[\frac{X_1+X_2+X_3}{3}\right] = \mu$ (3)

$\bar{X} = \frac{X_1+X_2+X_3}{3}$ Sample mean

As $\bar{X}$ consistent estimater of μ. Therefore using WLLN, $\frac{X_1+X_2+X_2}{3}$ is a consistent estimator

(d) $Var[\theta_1] = Var X_1 + Var X_2 + Var X_3 = 3\sigma^2$

$Var[\theta_2] = 4 Var X_1 + 9 Var X_2 + 16 Var X_3 = 29\sigma^2$ } (4)

$Var[\theta_3] = \frac{1}{9}(Var X_1 + Var X_2 + Var X_3) = \frac{\sigma^2}{3}$.

θ_3 is the best estimater of μ as the variance is minimum

Q2. Let $X_1, X_2, \ldots, X_n$ be a random sample with Distribution function F. Find moments of sample mean

$E[\bar{X}]$, $V[\bar{X}]$, $E[\bar{X}]^3$,

in terms of $m_k = E[X^k]$ and $E[(X-\mu)^k] = \mu_k$,

(kth order population moments).

Q2

$$E[\bar{x}] = \mu$$

$$V[\bar{x}] = \operatorname{var}\left[\frac{x_1 + x_2 \cdots + x_n}{n}\right] = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n} \quad (1)$$

$$E(\bar{x})^3 = E\left[\frac{1}{n}\sum_{j=1}^{n} x_j\right]^3 = \frac{1}{n^3} E\left[\sum_j x_j^3 + 3\sum_{j \neq k} x_j^2 x_k + \sum_{j \neq k \neq l} x_j x_k x_l\right] \quad (2)$$

Noting $m_k = E[X^k]$ and $E[(X-\mu)^k] = \mu_k$ for kth order population moments, we get:

$$E(\bar{x})^3 = \frac{1}{n^2}\left[m_3 + 3(n-1)m_2\mu + (n-1)(n-2)\mu^3\right]$$

Q3. Suppose we have five independent observations on $X \sim b(1, p)$ are 0, 1, 1, 1, 0.
Find the sample mean and sample variance

Q3. $\bar{x} = \frac{0+1+1+1+0}{5} = 0.6$

$$s^2 = \sum_{i=1}^{5} \frac{(x_i - \bar{x})^2}{5-1} = \frac{2(0.6)^2 + 3(0.4)^2}{4}$$

$$= 0.3.$$

Q 4 Let $X \sim N(\mu, \sigma^2)$ where μ is known and σ^2 is unknown. Let $X_1, X_2, \ldots, X_n$ be a sample from $N(\mu, \sigma^2)$. Is

$$\sum_{i=1}^{n} X_i / \sigma^2$$

a statistic?

Q4 Noting that $f(x_1, x_2, \ldots, x_n)$ is a statistic provided it is <u>not a function</u> of any unknown <u>parameter</u>. $\sum_{i=1}^{n} x_i / \sigma^2$ is not a statistic

Therefore

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Q5. Let $(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$ be a sample from a bivariate population with variances σ_1^2, σ_2^2 and covariance $\rho\sigma_1\sigma_2$.

Then show that

$$E S_1^2 = \sigma_1^2, \quad E[S_2^2] = \sigma_2^2$$

$$E[S_{11}] = \frac{1}{n-1}\left[\sum_{j=1}^{n}(X_j - \bar{X})(Y_j - \bar{Y})\right] = \rho\sigma_1\sigma_2.$$

Q5 We know that

$$E[S_i^2] = \sigma_i^2 \quad , i = 1, 2$$

Note X_i is independent of X_j $(i \neq j)$ and Y_j $(i \neq j)$.

$$(n-1)\, E[S_{11}] = E\left[(X_j - \bar{X})(Y_j - \bar{Y}) \right] \quad \text{--- (1)}$$

$$E[(X_j - \bar{X})(Y_j - \bar{Y})] = E\left[X_j Y_j - X_j \frac{\sum Y_j}{n} - Y_j \frac{\sum X_j}{n} + \frac{\sum X_j \sum Y_j}{n^2} \right]$$

$$= E[XY] - \frac{1}{n}\left[E(XY) + (n-1)\, E(X)\, E[Y] \right]$$

$$- \frac{1}{n}\left[E(XY) + (n-1)\, E(X)\, E(Y) \right]$$

$$+ \frac{1}{n^2}\left[n\, E[XY] + n(n-1)\, E[X]\, E[Y] \right] \quad (2)$$

$$= \frac{n-1}{n}\left[E[XY] - E(X)\cdot E(Y)\right]$$

or $(n-1)\,E[S_{11}] = n\,\frac{n-1}{n}\,E\left[XY - E(X)E(Y)\right]$

or $E[S_{11}] = E(XY) - E(X)E(Y) = \mathrm{cov}(X,Y)$

$= \rho\sigma_1\sigma_2$